Acoustic Analysis Progress

Methodologies and Techniques applied to the dataset Aziz et al., 2025 and their results

- This dataset includes inner fault, outer fault, and normal condition with 2160 files recorded at 10-second intervals.
- Since we don't have historical data, we will stick to traditional signal processing techniques.
- In the time domain, vibration signals from normal and faulty bearings appear visually similar, making fault identification difficult without advanced signal processing
- Our goal of finding fault detection, be it inner fault or outer fault. Upon exploring all possible signal processing techniques. We finalized a few that could be relevant to our bearing experiment.
- We started with most fundamental signal processing technique, a statistical method known as kurtosis analysis, but kurtosis over the whole signal failed since it identifies impulsiveness over the whole signal but fault impulses are often localized on particular frequency
- The next obvious method was to apply kurtosis analysis over particular bands, which is known as spectral kurtosis. Before applying spectral kurtosis, we explored another signal processing technique known as Wavelet decomposition
- In the following slides we will show each technique's implementation details and the reasoning behind it and the results obtained.

Wavelet Decomposition

- Wavelet decomposition, by principle, works by splitting the signal into multiple frequency bands and then applying low-pass and high-pass filters concurrently. We will consider 5 files per condition for our initial experimentation ,

```
import numpy as np
import pywt  # PyWavelets library

# Configuration
WAVELET_TYPE = 'db4'  # Daubechies 4 wavelet
DECOMPOSITION_LEVELS = 6  # 6 levels = 7 frequency bands
SAMPLING_FREQ = 25600  # Hz
```

- d4 was chosen as the main wavelet for our analysis because this wavelet shape closely mimics these mechanical impulses. A higher correlation between the wavelet and the signal results in higher coefficients, making the fault easier to detect.
- We have assumed 6 levels are enough to capture relevant frequency {200-12800 Hz}
- We have assumed normalized energy distribution since our goal is to find fault detection and fault the distribution pattern, not absolute energy. This method will help us finding a particular frequency band having highest discrimination.

- Once the conditions are set , we run wavelet decomposition on each signal of all conditions.
- We will get normalized energy levels for all conditions. Assuming normal condition as a baseline, we can get inner race fault and outer fault as relative percentages.

Level	Freq (Hz)	Normal	Inner	Outer	IR vs N	OR vs N
D1	6400–12800	63.03%	61.53%	64.24%	−2.4%	+1.9%
D2	3200–6400	17.90%	18.35%	13.18%	+2.5%	−26.4%
D3	1600–3200	10.44%	10.62%	10.97%	+1.7%	+5.1%
D4	800–1600	5.23%	5.50%	6.61%	+5.2%	+26.4%
D5	400–800	1.99%	2.39%	3.09%	+20.1%	+55.3%
D6	200–400	1.39%	1.59%	1.89%	+14.4%	+35.9%

- *Normal / Inner / Outer: Fraction of total signal energy in each wavelet band (energies sum to 100% per condition); e.g., D2 = 17.9% means 17.9% of the energy lies in 3200–6400 Hz.*
- *IR vs N / OR vs N: Relative change in band energy compared to Normal, computed as ((Fault - Normal) / Normal) * 100); e.g., D2 OR vs N = −26.4% means the outer-race condition has 26.4% less energy in that band than Normal.*
- The D2 band has the biggest gap if compared by absolute value, while D4 and D5 have the biggest percentage when compared to normal. While running the fault detection algorithm during D5 and D6, the fault impulse might here appear significant.

Evaluation on different dataset (Mendeley dataset)

- The same evaluation as was done previously was done on this dataset. With **db4** as the mother wavelet and a sampling frequency of **44.1 kHz.**
- In this particular dataset, we had data for faults of around 600 rpm, and 1800.
- running wavelet decomposition on both rpm led to the following result:

600 RPM

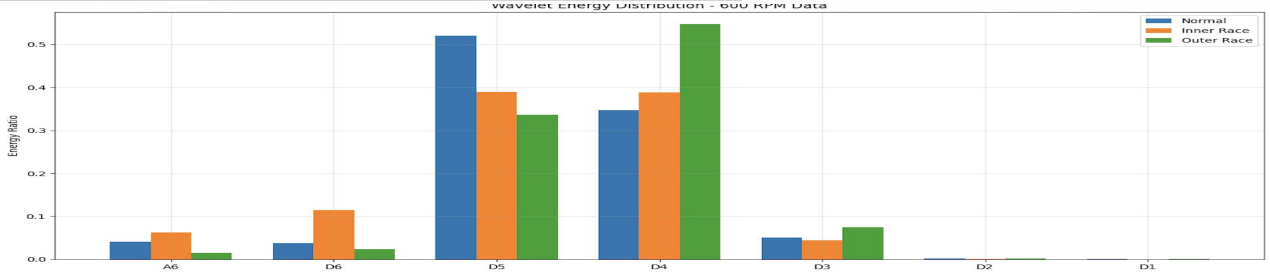
Band	Normal	Inner Race	Outer Race
A6	4.13%	6.27%	1.53%
D6	3.75%	11.42%	2.34%
D5	**52.08%**	**38.96%**	33.65%
D4	34.70%	38.81%	**54.79%**
D3	5.02%	4.46%	7.46%
D2	0.21%	0.08%	0.19%
D1	0.11%	0.00%	0.04%

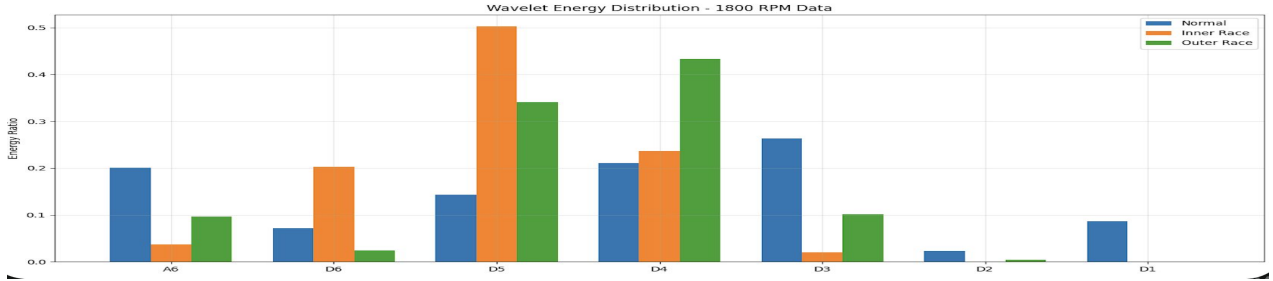
Here, percentage signifies energy distribution.

1800 rpm

Band	Normal	Inner Race	Outer Race
A6	20.11%	3.73%	9.63%
D6	7.21%	**20.29%**	2.42%
D5	14.33%	**50.24%**	34.08%
D4	21.08%	23.65%	**43.27%**
D3	**26.29%**	2.01%	10.12%
D2	2.29%	0.07%	0.46%
D1	8.68%	0.01%	0.02%

Here percentage signifies energy distribution





Dominant Band across all conditions

Dominance Strength

Energy Spread

High Frequency activation

Results and conclusion from wavelet decomposition

- As shown previously , D5 and D6 could be used for early detection of faults.
- A very important result we got is that fault detections are primarily from outer fault conditions. **The chance of an inner fault being detected** is very low, as seen from the table.
- Wavelet decomposition works reliably well for outer fault detection, but it might miss inner fault defects.
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Spectral Kurtosis

- This leads us to spectral kurtosis, where we apply kurtosis on a particular frequency band.
- For this signal processing technique we will consider one normal file , and consecutive inner and outer faults for our experimentation.
- For this particular analysis, we attempted implementing this paper Spectral kurtosis on AE signal
- Initially we took one normal condition file , and 3 consecutive inner fault and outer fault files. But it might not be enough for detecting the correct fault, as it showed a fault at normal condition, which might be due to some machine component . So we took 10 consecutive normal inner faults and outer faults .
- According to this paper, spectral kurtosis surpasses wavelet decomposition in detecting faults.
- We start by plotting a kurtogram - a 2D spectral kurtosis map varying both frequency and window size and then use it to find a particular frequency band
- From the kurtogram we found the 1000-1400 Hz band , we will apply a bandpass filter here to eliminate noise and then implement envelope analysis on this particular band.
- Dataset we con
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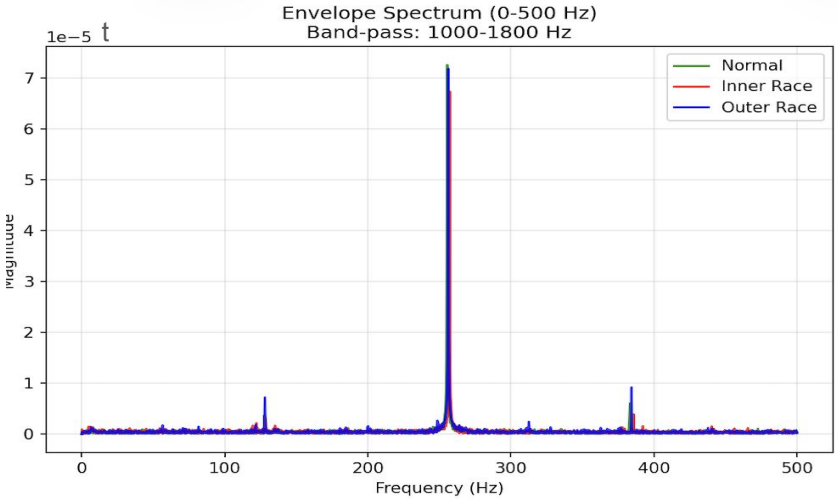
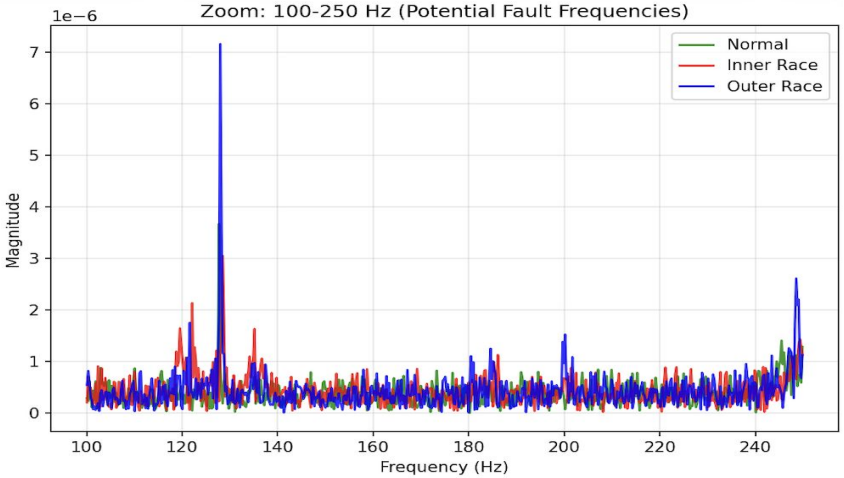
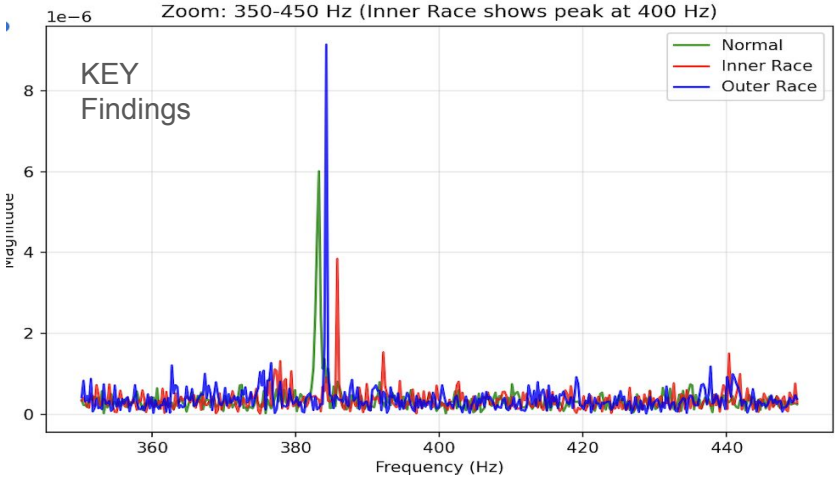
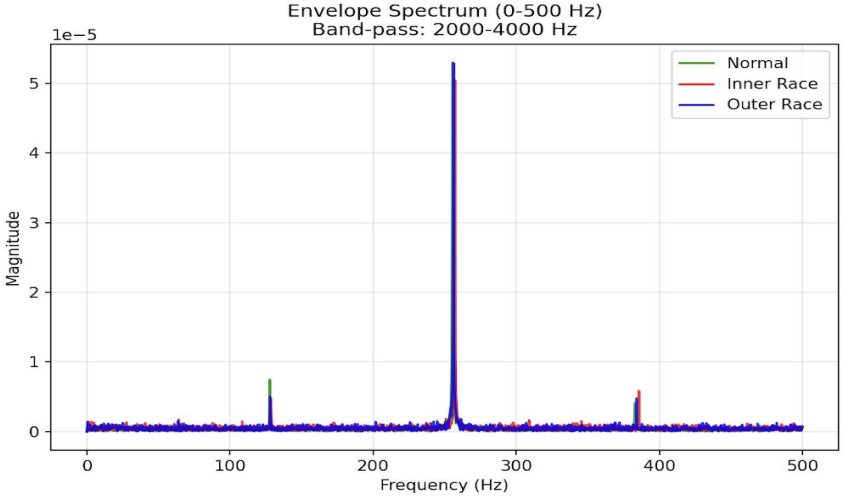
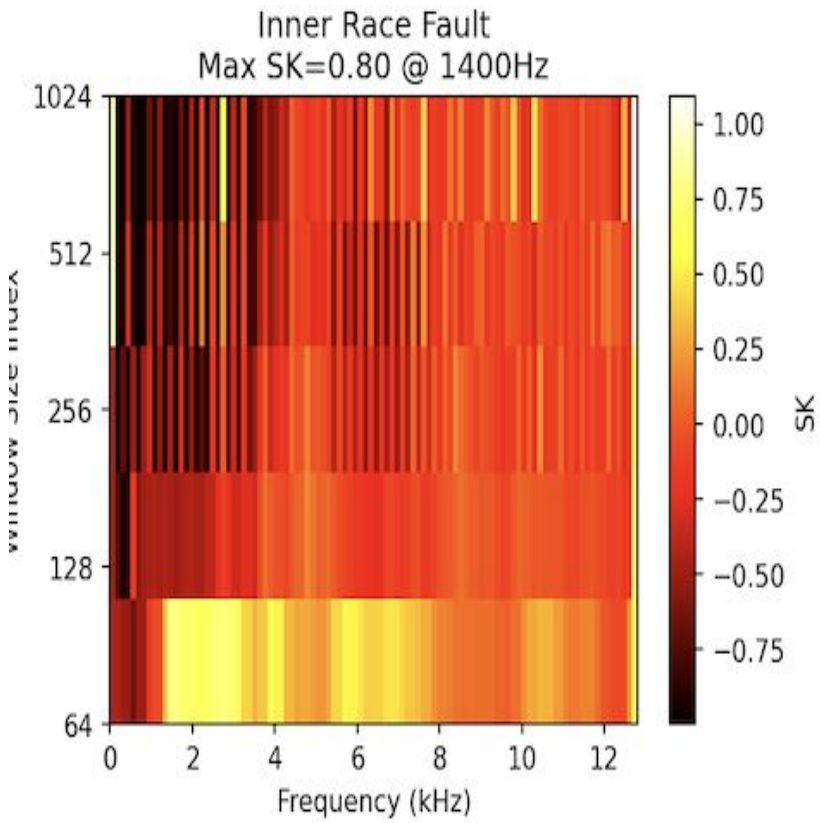

Envelope Spectrum (0-500 Hz)
Band-pass: 1000-1800 Hz
1e−5
Normal
Inner Race
Outer Race
Magnitude
Frequency (Hz)

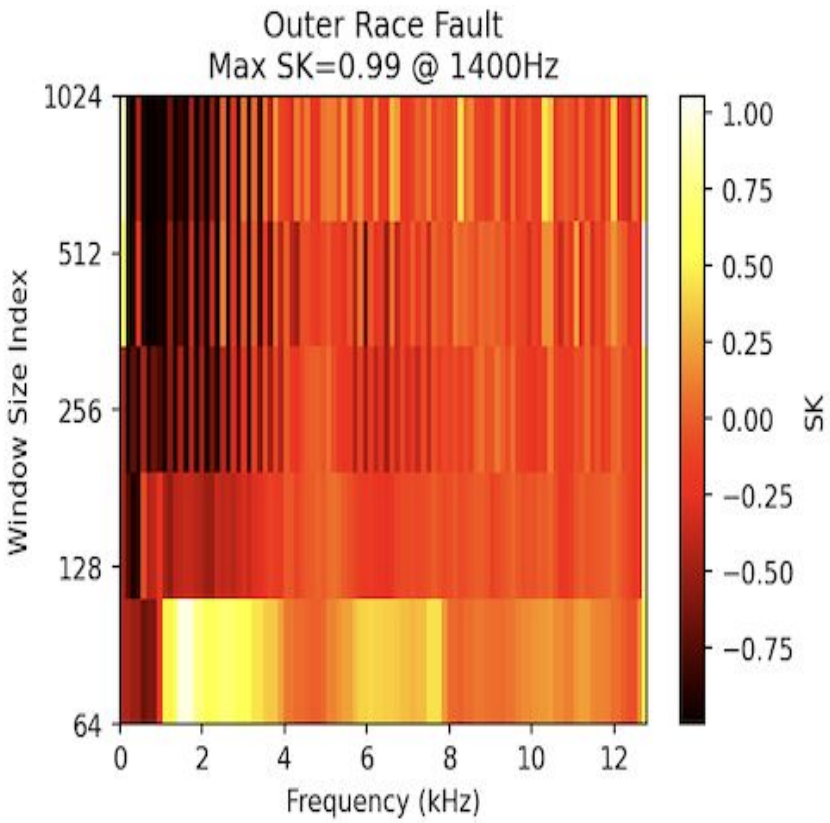

Zoom: 100-250 Hz (Potential Fault Frequencies)
1e−6
Normal
Inner Race
Outer Race
Magnitude
Frequency (Hz)


Zoom: 350-450 Hz (Inner Race shows peak at 400 Hz)
1e−6
KEY
Findings
Normal
Inner Race
Outer Race
Magnitude
Frequency (Hz)


Envelope Spectrum (0-500 Hz)
Band-pass: 2000-4000 Hz
1e−5
Normal
Inner Race
Outer Race
Magnitude
Frequency (Hz)

Kurtogram Plot





Experimentation on the Mendeley Dataset

- Our initial observation on the dataset Aziz et al . 2025 .Max spectral kurtosis was very low for both inner fault and outer fault, as shown on Previous Slide which would make it hard for fault detection.
- On this dataset, which had data at around 600 rpm and 1800 rpm .We will start by calculating maximum spectral kurtosis for each fault condition and then subsequently plot kurtogram for identifying suitable band where we could do envelope analysis
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